Diagnosing and Repairing Out-of-Distribution Failure in MOBA Draft Policies

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***Abstract*—In multiplayer online battle arena (MOBA) drafting, two teams alternately ban and pick heroes from a shared pool to build competing five-hero compositions, and a noisy match decides the winner. In Heroes of the Storm, a 90-hero pool spans 7.2×10^{14} matchups; 2.1 1.94 million ranked drafts cover a vanishing, clustered fraction, so out-of-distribution behavior dominates policy quality. Search exploits learned value functions' out-of-distribution optimism, assembling structurally broken teams rated as competitive, and pessimistic offline reinforcement learning does not repair this: conservative and implicit Q-learning collapse to behavioral cloning, and mildly conservative Q-learning to a near-deterministic ten-hero policy. We decompose draft-policy quality into composition safety and interaction-awareness, with metrics that detect both collapses, and show that role-based heuristics over 9 million community games identify never-observed compositions as reliably bad. A layered repair (domain features , targeted synthetic data , . We also report a pitfall in our own first pipeline: features built from aggregate statistics that contained each game's own outcome made the value function overconfident and inflated its reported accuracy and gains. With out-of-fold statistics, and with judges that share no data with any agent and are calibrated on real broken teams, the repair that works is search plus a structural mask. Monte Carlo tree search , structural constraint masking) guarantees validity by construction, preserves the interaction-awareness that behavioral anchoring destroys, and wins every tournament matchup at 0.67 consensus win probability, an orderingreproduced by judges trained on a draft-free game modeover a value function with domain features is interaction-aware, and a mask that removes any pick leaving no valid completion eliminates the broken teams search still builds, with a small gain in win probability. Search beats every behavior-anchored method under every independent judge (0.61 vs. 0.44–0.46 tournament win probability). Extra training simulations and domain features raise the agents' own value estimate but not their independent win probability; the features buy composition safety instead. A registered expert study could not serve as a further check: nine experienced players agreed with our value functions at chance (50% of 488 judgments) and preferred the behavior-anchored drafts, but they also picked the winners of real mid- and high-tier games at chance, so their judgment neither confirms nor refutes the ordering, which rests on outcome-based checks. Code, models, and the in-browser agent are open-source.**

***Index Terms*—MOBA drafting, team assembly, offline reinforcement learning, Monte Carlo tree search, out-of-distribution generalization.**

I. Introduction

Drafting has a structure with three properties that drive everything in this paper: a small number of opponents (here, two) two opponents alternately claim discrete, rivalrous elements from a shared pool to assemble competing sets; the value of a set depends nonlinearly on interactions within it and against the opposing set, not only on element quality; and the winner is determined by ; and a noisy downstream process rather than by rule. The same structure recurs wherever rivals assemble from a shared scarce pool (sports drafts , roster auctions, deck drafting, procurement), but those instances lack instrumentation; drafting decides the winner. Sports drafts share this structure but are rarely logged at scale. Drafting in multiplayer online battle arena (MOBA) games offers millions of logged episodes under fixed rules, a natural model organism. In Heroes of the Storm (HotS), two teams of five alternate through 16 actions, banning 6 heroes and picking 10 from a pool of 90, before the match itself is played. We study HotS because it HotS is in maintenance mode: a fixed 90-hero pool across 4.5 years of one major version yields an unusually clean gives an unusually stationary corpus. The draft matters: models that observe see only the completed draft predict match outcomes at 56–58% accuracyoutcomes at about 57% on held-out games, a large edge in a balanced competitive game.

Two properties of the domain make learned drafting hard. The first is coverage. A 90-hero pool admits : roughly 7.2×10^{14} distinct final matchups ($\frac{1}{2}\binom{90}{5}\binom{85}{5}$), and our corpus of 2.1 million ranked drafts covers a vanishing fraction of that space, and not uniformly: human play clusters around popular heroes and familiar groupings, leaving the rest not merely sparse but empty. Any drafting policy must therefore of which our 1.94 million drafts cover a vanishing and clustered fraction, so a policy must behave sensibly in states it has effectively never observed. The second is evaluation: unlike chess or Go, where the terminal state determines the winner by rule, a completed draft only shifts the odds of the matchthat follows, so the reward signal for training and search is itself a win probability model estimated from the same sparse , clustered data, and value and policy errors compound.

Prior work has treated

Prior work treats draft outcome prediction and hero recommendation largely as supervised learning problems and reports respectable and reports held-out accuracy [4], [5], [7], [6]. We show that this That accuracy is a misleading foundation for a drafting policy poor guide to policy quality. Three value functions separated by less than half a percentage point of test accuracy (57.7–58.1within 0.2 percentage points of held-out accuracy (57.5–57.6%) yield greedy policies that disagree on roughly 90% of decisions. When used for different greedy policies; the naive and enriched policies make the same pick at matched states only 7–13% of the time. Under search, each model's out-of-distribution behavior , not its accuracy, determines behavior out of distribution decides policy quality: the optimizer actively seeks out states finds states that are vanishingly rare in human play, a team of five tanks , or one with such as five tanks or no healer, and the models score them as competitive. The evidence that such teams fail exists, but not in any individual game: lineups like these are essentially absent even from community aggregate statistics covering 9 million ranked games. Only is visible only when heroes are grouped by roledoes the pattern become visible: the role-degenerate compositions players do occasionally field consistently underperform. Directly

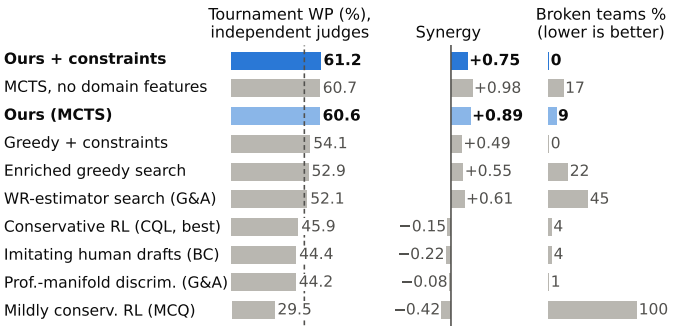


Fig. 1. Results at a glance: the eleven-strategy round-robin (110 ordered pairs × 200 drafts; Section VII-A). First column: consensus mean win probability over the under four evaluators composition-corrected judges that share no data with any agent (dashed line: 50%); the CQL row shows the better variant per column(both in Table IV). Anchoring to human behavior is safe but interaction-blindblind to interactions; unstructured greedy value search is interaction-aware sees interactions but is unsafe; only domain features + MCTS achieves all threereaches both, and constraint masking (Section VII-B) removes its residual broken teams and tops the tournamentwith a small gain in win probability.

exploiting this evidence requires representing roles, domain structure that hero-identity inputs do not carry.

The offline reinforcement learning literature offers a standard remedy: standard offline RL remedy is pessimism toward out-of-distribution actions [1], [9]. Applied to drafting, where reward is terminal and With terminal reward and exact Monte Carlo returnsare exact, algorithmic pessimism degenerates reduces to behavioral anchoring by construction (Section VI), and its two instantiations collapse in characteristic ways. Conservative Q-Learning (CQL) appears to succeed, reducing cuts degenerate compositions to 2–3% , but richer evaluation reveals it does so 1.5–3.6% by collapsing to behavioral cloning : 69–74(68–73% agreement with the behavior policy, a concentrated pick distribution, and negative synergyexploitation. negative synergy), and Mildly Conservative Q-Learning (MCQ) collapses differently, to a near-deterministic policy over 10 heroes. We decompose draft policy ten-hero policy. We therefore split draft quality into two separable axes: axes. **composition** Composition **safety** , whether the team includes asks whether a team has the structural roles it needs (healer, frontline, ranged damage)required to function at all, measured by healer rate and degenerate composition rate; and . **interaction-awareness**Interaction-awareness , whether the team asks whether it responds to the opponent 's picks and exploits internal synergies, measured by counter responsiveness, resilience to subsequent counter-picks, synergy exploitation, and pick diversity. Safety turns out to be . Safety is the easy axis, inherited by any method that stays near human behavior; interaction-awareness is the hard axis, and no pessimism-based method we tested achieves itinteraction-awareness.

What achieves both is domain structure amplified by search : value function features built from the role-aggregated statistics (role counts, pairwise counter and synergy rates, composition win rates), search plus a structural mask. AlphaZero-style Monte Carlo tree search (MCTS) over the resulting value function a value function with features built from role and outcome statistics finds interactions, and a final hard mask that prunes structurally invalid picks by rule .

Fig. 1 previews the outcome: in an eleven-strategy head-to-head tournament, only the combination of domain features removes the broken teams search still builds. Value functions misjudge broken teams because training data almost never contain them (Section V), and a rule can describe them exactly. Building those statistics correctly matters. In our first pipeline they came from community aggregates that contained each game's own outcome, deep search, and structural constraint masking wins every matchup while remaining both safe and interaction-aware. Its margin holds under evaluators sharing no features with its value function (0.657/0.666 test games included. The leak inflated held-out accuracy, made the value function overconfident, and inflated its own measure of the gains from features and simulations. Independent judges confirm neither the feature gain nor the simulation gain, and removing the leak does not change that; with the leak, more search makes the agents worse under those judges (Sections III-B and VII). This revision rebuilds every value function on out-of-fold statistics and judges every agent with references that share no data with it. In an eleven-strategy tournament judged that way (Fig. 1), MCTS agents lead at 0.61 win probability, naive/hero-strength evaluators), and its orderingis reproduced by judges trained on Quick Match, a draft-free game mode whose corpus no model here ever touched (Spearman 0.92, both a 2021-era and a current-era judge): the lead is not self-gradedgreedy searchers follow at 0.52–0.54, and every behavior-anchored method trails at 0.44–0.46. We also tested whether experienced players could check this ordering. They could not: they agreed with the models at chance, but they were also at chance on the winners of real games above the lowest tier (Section VIII). Our contributions:

1) A decomposition of draft-policyquality into composition safety and interaction-awareness, with domain-agnostic metrics (counter responsiveness, resilience, synergyexploitation, pick diversity, behavioral similarity) that distinguish genuine separate compositional reasoning from mimicry. Using these, With them we show that CQLachieves , BC-regularized CQL, and IQL reach safety by behavioral cloning , robustly across architectures and feature sets, and and that MCQ's penalty does not transfer to this discrete setting, collapsing to a near-deterministic ten-hero policy.
2) Domain-specific aggregate heuristics : role-composition win rates from community aggregate statistics spanning 9 million games, a corpus separate from and larger than our drafts, that confidently identify Role-composition heuristics that flag classes of compositions as bad despite their absence from individual-game data . These heuristics near-absence from game data (no-healer teams win 42% in our corpus), used to evaluate policies, populate value function build features, and target synthetic training data at never-observed states. define a structural mask. We show that such statistics must be computed out of fold: in-sample statistics leak each game's outcome into its own features, and the value function becomes overconfident (calibration slope 0.75 vs. 0.96).
3) A layered repair, enriched features, targeted synthetic data, MCTS A repair that reaches both axes: MCTS self-play , and structural constraint masking, that achieves both axessimultaneously, cutting degenerate compositions from 64% under naive value-guided search to 8%, with the final mask guaranteeing structural validity by construction, and with the enriched value function contributing +0.023 average win probability over an identical

agent without domain features($p < 0.001$); data scale narrows this gap under deep search, while features remain critical for shallow search and OOD behavior. Empirically, validity masking preserves interaction-awareness (synergy +0.95) where behavioral anchoring destroys it, and the constrained agent beats its unconstrained counterpart head-to-head, winning all 20 tournament matchups against every baseline familyover a value function with domain features, plus a structural mask. Under independent judges, MCTS beats every anchored and greedy baseline, including reimplementations of an independently designed a published drafter and of an earlier win-rate estimatorby its lead author. . At matched search, features leave independent win probability unchanged and cut degenerate teams from 15.4% to 4.5%. The mask guarantees 0%, keeps positive synergy, and adds about 0.01 win probability, and the masked agent places first in the tournament.

4) An open-source systems stack:[1] a fused CUDA MCTS implementation sustaining 140–190 kernel sustaining 140–161 episodes/second per GPUdepending on multi-GPU contention, 2.2× a tuned batched-evaluationhost pipeline at equal search quality (supplement), and a an in-browser deployment of the complete agent , search, policy, value, and opponent models, running in-browser at interactive latency full agent in a ∼20 MB download.

II. Background and Related Work

A. *Drafting in MOBA and Esports Games*

Drafting. Game-playing agents for MOBA and real-time strategy games have concentrated focus on in-match controlrather than the draft. OpenAI Five [13] reached professional level in restricted Dota 2 but restricted play to 17 heroes , drafting and drafted by exhaustive searchover the reduced (∼5M) matchup space; AlphaStar [14] faced no draftstage at all; at full pool sizes the draft becomes a distinct combinatorial game that exhaustive methods cannot touch.

has no draft. Draft-specific work falls into two camps. Outcome prediction estimates win probability from completed or partial drafts and optionally searches against it: Chen et al. [4] used such a predictor as the value function for MCTS-based recommendation MCTS over a predictor in Dota 2, and JueWuDraft [8] scaled the recipe to Honor of Kingsbest-of-N series. Recommendation by imitation learns to reproduce . Imitation reproduces expert selections: Summerville et al. [7] predicted professional Dota 2 draft picks with sequence models; , Gourdeau and Archambault [5] trained a discriminatorscoring a draft's proximity to the professional-play manifold in HotS and Dota 2, drafting to minimize divergence from it; drafted by minimizing a discriminator's distance to professional play, and DraftRec [6] personalizes suggestions while restricting them to observed hero combinations. All three imitation approaches sidestep restricts recommendations to observed combinations. Imitation sidesteps out-of-distribution states by construction, at the cost of novel strategydiscovery. Gourdeau and Archambault also conjectured that MCTS over a win-rate estimator would wander into compositions poorly characterized by low-density data, proposing to constrain such poorly characterized compositions and proposed constraining search to the professional manifold. We empirically confirm the conjectured confirm the failure (Section IV), show that behavioral anchoring , the remedy family their proposal belongs to, achieves composition safety but not anchoring gives safety without interaction-awareness (Section VI), and evaluate an independently released win-rate estimator by evaluate the same authoras an additional baseline 's released win-rate estimator (Section IV-0a). Section VII-B implements a structural-validity version of their constraint proposal on top of our learned agent, and implement a structural version of the constraint (Section VII-B).

A. *Offline RL and Pessimism*

Offline RL addresses out-of-distribution value overestimation with pessimism, applied at one of two loci. Algorithmic pessimism penalizes value estimates directly: Conservative Q-Learning (CQL)*Offline RL and pessimism.* CQL [1] pushes down Q-values for of actions underrepresented in the data, and Mildly Conservative Q-Learning (MCQ); MCQ [9] penalizes only Q-values exceeding a threshold, targeting CQL's tendency toward over-conservatism. Policy constraint keeps the learned policy near the behavior policy: above a threshold; BEAR [3] via distributional matching, and TD3+BC [15] via an explicit behavioral-cloning term; DraftRec's restriction to observed combinations is a domain-specific instanceconstrain the policy toward the behavior policy; IQL [10] avoids querying unseen actions. Levine et al. [2] survey the area. Our work adds a third locus: pessimism expressed We add structure through features and data, with no algorithmic modification; Section VI shows that CQL, MCQ, and BC-regularized CQL each collapse, in different ways, before reaching interaction-aware behaviora rule-based mask instead.

A. *Value Functions Under Search*

That search amplifies both the strengths and *Value functions under search.* Search amplifies the errors of its value functionis well established in games [11], [12], reappearing as reward model as in reward-model overoptimization [16]; our setting differs in that no ground-truth terminal evaluation exists even in principle, here no terminal evaluation by rule exists, and the value function being is learned from the same sparse data the policy must generalize beyond.

A. *Scaling MCTS*

MCTS parallelization classically distributes tree operations across CPU threads with batched or asynchronous network evaluation, root, leaf, and tree parallelism [17], and the GPU-batched self-play pipelines of AlphaZero [12] and KataGo [18], keeping the tree on the host and paying a host-device round trip per evaluation batch. Our fused CUDA implementation instead colocates the entire Features built from outcome statistics add target leakage [19]: when a row's features include its own outcome, the model over-trusts them and is overconfident wherever the statistics lack the outcome, which includes every state search generates. The standard remedy is out-of-fold or ordered statistics [20]. Our CUDA kernel differs from host-tree MCTS pipelines [17], [18] by running the whole episode, tree and network, inside a single kernel launch, suited to the draft domain's small networks and short episodesnetworks, in one kernel launch.

III. Problem Setup

A. *Draft Formalization* and Data

A HotS draft consists of draft is 16 sequential actions : actions (6 bans(removing heroes from the pool) and , 10 picks(selecting heroes for teams)

[1]https://anonymous.4open.science/status/moba-draft-policies

, alternating between two teams) in a fixed order. The state s_t at step t comprises holds multi-hot vectors for of each team's picks and combined bansthe bans, map ($\in \{0,1\}^{90}$ each), map identity (one-hot over 14maps), skill tier (one-hot over 3tiers), stepnumber, and step type(ban or pick), for : 289 dimensionstotal. The action space is the set of available heroes (∼84–90 early, decreasing as the draft progresses); terminal . Terminal reward is the game outcome.

The behavior policy π_β is defined by our corpus of ranked Storm League replays(2.07 million drafts, continuously collected). All experiments use a snapshot , pinned in May 2026 and filtered to the current major patch (patch 2.55): 1,949,087 replays spanning 943,338 replays from December 2021 through to May 2026, a window over which with the hero pool is fixed at 90. Replays cover the full skill range, collapsed to three tiers (Bronze–Gold, Platinum–Diamond, Master–Grandmaster). These ranked drafts arise largely from uncoordinated Skill enters as three tiers from each game's average rank: low is Bronze and Silver (41.1% of games), mid is Gold and Platinum (41.2%), and high is Diamond and Master (17.7%); games with no recorded rank are excluded. These drafts come mostly from solo and duo queue, where players pick by hero preference and comfortrather than explicit win maximization, so π_β reflects social dynamics habit as much as strategic intentstrategy. All models use team-swap augmentation. Splits are at the replay level: the test set is the submission's test replays (38,869 with a recorded rank), a 2% hash-selected validation set (38,swapping teams and all team-indexed features with flipped outcome, doubling the effective training set to 3.9M samples. Train/test splits are at the replay level throughout: step-conditioned models generate ∼10 correlated partial states per game,so sample-level splitting would leak game identity and inflate held-out accuracy (∼8pp). 100) serves early stopping and seed selection, and 1,866,369 replays train (3.73M swap-augmented rows). Every model in this paper, baselines included (the Gourdeau estimator, GD, CQL, IQL, MCQ, and the discriminator), is trained on this split and selected on validation games, never on test data.

a) Post-snapshot games*:* Our database kept collecting after the snapshot. We hold out 260,539 later games that no agent, value function, or agent statistic touched; they are used only to build the independent references (Section III-D): 136,517 snapshot-era games uploaded after the snapshot and 124,022 on the build in force at snapshot time (2.55.16.97039), with no balance change. A further 154,387 games on the 2.55.17 builds, after a balance patch, serve as a drifted check. No game from a build released after 2026-09-27 is used.

B. *Value Function Variants*Functions and Outcome Leakage

Table I shows three win probability models trained on the same data, plus the reproduced baseline. The lists the win probability (WP) models. The naive model sees hero multi-hots, map, and tier; hero strength adds per-hero win rates; the enriched model adds 86 features: per-team counts of 9 fine-grained roles role counts (18), per-hero counter deltas against the opposing team opponents (50), team-levelpairwise counter and synergy aggregates (4), team-average and map-specific win rate map win-rate deltas (4), meta strength (4), draft diversity (2), and empirical composition win rates (4). The composition WR feature maps the empirical win rate of each team's 5 heroes to a sorted role tuple and looks up its empirical win rate from aggregate statistics published by a community replay database ,[2] covering 9.0M games across three skill tiers (164/127/121 distinct compositions per tier, of 252 possible). Over 99.9% of training replays match a recorded composition; the sorted role composition (4), with a pessimistic 33% pessimistic default fires for degenerate compositions that humans rarely draft. These published aggregates are a second data source, distinct from our replay corpus: several times more games, but only per-hero, per-pair, and per-role-composition win rates, with no individual drafts; all role-level heuristics in this paper derive from them. % default for compositions below 50 games.

TABLE I

WIN PROBABILITY MODEL VARIANTSmodels. THE TOP THREE USE IDENTICAL ARCHITECTURE Submitted: the original checkpoints and external statistics. Leak-free: retrained on out-of-fold statistics from our training split, early-stopped on validation; mean of 3 seeds (SD ≤0.14pp). Test: the 38,869 held-out replays. Post: 260,539 post-snapshot games. Slope: calibration slope on Post (1 is calibrated, below 1 overconfident; 3-seed SD 0.01–0.03). On Test every leak-free model's slope exceeds 1 (1.03–1.22, naive and the Gourdeau estimator included), a property of that sample, so we use Post. All rows 256→128 MLP, DROPOUT 0.3) AND TRAINING (ADAMW, COSINE LR, EARLY STOPPING, TEAM-SWAP AUGMENTATION)AdamW; GOURDEAU REPRODUCES IS A SEPARATELY RELEASED designed ESTIMATOR (SECTION IV-0A), retrained on our split.

	Submitted			Leak-free			
Model	Test	Post	Slope	Test	Post	Slope	Dim
Naive	57.6	56.8	1.03	57.6	56.8	1.08	197
Hero Str.	57.6	56.6	1.01	57.5	56.5	1.02	209
Enriched	58.0	56.7	0.89	57.5	56.8	0.96	283
Enr., in-sample	–	–	–	57.2	56.3	0.75	283
Gourdeau	–	–	–	56.5	55.7	0.91	194

The leak. The submission computed these statistics from aggregates published by the community database our replays come from. Our replays make up a median 36–79% of the games behind each hero statistic, so every game's features, test games included, contained its own outcome, most strongly in sparse pairwise cells. Removing each test game's own contribution drops the submitted enriched model from 58.0% to 56.9%; removing as many random training games changes nothing. Trained on rows whose features contained their own labels, the model learned to over-trust the pairwise features, and on games outside the statistics it is overconfident (post-snapshot slope 0.89 vs. 1.03 for the naive model).

The fix. We now compute every statistic, including the composition table, from our own training split, out of fold: training rows in each of five hash folds get features from the other four folds; validation, test, post-snapshot rows, and every searched draft use statistics over the whole training split, which contain none of them. The external aggregates no longer enter any value function or judge trained for this revision; one inherited tournament baseline (CQL, enriched) still reads them. The leak-free enriched model is nearly calibrated on unseen games (slope 0.96) and as accurate as the naive model (56.8% post-snapshot; 57.5% vs. 57.6% on test). The same model with in-sample statistics loses half a point and is strongly overconfident (slope 0.75). The submitted enriched model's 0.3pp edge over the naive model on test

[2] heroesprofile.com

came from the test games' own outcomes; post-snapshot it has none. We report held-out calibration slope for every value function and judge.

What the leak does to search. To isolate the mechanism without balance drift we split the snapshot by hash into halves A and B, trained value functions on A, and judged search against them with references built only from B (supplement). With in-sample statistics a value function trained on 952K games reports 59.7% on its own validation rows, reaches 56.2% on B, and has slope 0.62; with out-of-fold statistics it reports 56.2%, reaches 57.0%, and has slope 1.02. Raising search from 1,024 to 16,384 simulations lifts the in-sample value function's own score by 0.080 and the B references by 0.027; the out-of-fold value function's own score (+0.060) and the B references (+0.061) move together. A small in-sample value function (119K games) shows classic over-optimization: its own score climbs to 0.94 while the B references stop improving after 4,096 simulations. In self-play the gap is starker: from 200 to 800 training simulations the in-sample agents lose 0.079 under the B judges and field 75% degenerate teams, while the out-of-fold agents gain 0.016. Calibration slope flagged the problem; the disagreement of a seed ensemble sharing the same statistics did not.

C. Evaluation*Framework*

We evaluate along two separable axes.

a) Composition safety.: Healer selection rate , and degenerate composition rate: missing healer, missing rate and degenerate rate: no healer, no frontline, no ranged damage, or 3+ heroes in one role. These four criteria are not stipulated; they are the composition classes that are the classes that role-composition aggregates identify as reliably underperforming , either falling well below 50% win rate where fielded statistics show underperforming (no-healer compositions win 43% of real games teams win 42.4% of 51,090 games in our training split) or absent from playentirely.

b) Interaction-awareness.: We measure interaction-awareness with complementary signals spanning two categories.

Interaction metrics (snapshot-dependent): both build on pairwise deltas , observed win rate minus its independence expectationfrom the heroes' individual win rates (in percent)Pairwise deltas are observed win rates minus their independence expectation: the counter delta of a against b is $\mathrm{WR}(a \text{ vs } b) - [\mathrm{WR}(a) + (100 - \mathrm{WR}(b)) - 50]$, and the synergy delta of teammates a, b is $\mathrm{WR}(a \text{ with } b) - [50 + (\mathrm{WR}(a) - 50) + (\mathrm{WR}(b) - 50)]$. *Counter responsiveness* , the average normalized pairwise counter delta against the is $\mathrm{WR}(a \text{ with } b) - [\mathrm{WR}(a) + \mathrm{WR}(b) - 50]$. *Counter responsiveness* averages our heroes' counter deltas against the opponents, *synergy* averages within-team synergy deltas, and *resilience* scores each pick against the opponent's heroes (positive = favorable). *Resilience*, a temporal complement: for each of our picks,the average counterdelta of the opponent's *subsequent* picks against it, negated: high resilience means picking heroes the opponent cannot profitably answer with their remaining picks. *Synergy exploitation*, picks, information no agent sees when choosing. Behavioral signals need no statistics: distinct heroes, pick entropy, top-10 concentration, and agreement with the behavioral-cloning model (GD similarity). The interaction metrics use the community aggregates as an evaluation instrument; no value function in this paper reads them. The two sources overlap in games, so some circularity between features and metrics remains. On 260,539 post-snapshot games, which the aggregates cannot contain, both metrics predict outcomes monotonically after controlling for team strength (+8.8pp synergy, +9.2pp counter, top vs. bottom decile; supplement).

D. Independent References

An agent cannot be graded by the value function it was trained or searched against. The submission's MCTS "Avg WP" and two of its four tournament judges did exactly that. The revision scores every generated draft with references that share no game, statistic, or search with any agent. *gN*: the average normalized within-team synergy delta (positive = heroes that win together more than their individual win ratespredict). Comparisons across strategies remain fair because all are evaluated identically, but absolute magnitudes are snapshot-sensitive (they shifted systematically when we re-pulled statistics), so we rest the collapse diagnosis on relative gaps and on the stats-independent signals below mean of three enriched judges trained with out-of-fold statistics only on the post-snapshot games, including a role-composition table built from those games. (A first version read the external composition table, which overlaps our corpus; the supplement compares the two.) *gN-naive*: a hero-identity judge on the same games. *RN*: a model-free realized-outcome index cross-fitted on the same games (hero win rates, synergy and counter residuals, and composition win rate from one half, weighted by a logistic fit on the other half). *QM2026*, *QM2021*: judges trained on Quick Match, a mode with no draft phase (116,589 and 91,365 games). On our test games, which none trained on, their held-out accuracy and calibration slope are 55.4%/0.92 (gN), 55.7%/0.95 (gN-naive), 54.7%/1.04 (RN), 54.0%/0.55 (QM2026), and 54.7%/0.40 (QM2021). The Quick Match judges are overconfident on ranked drafts, so we read them for ordering. Each reference has a blind spot: gN shares our model class, RN cannot judge compositions nobody plays, and the QM judges have never seen a draft. The evidence is their agreement. The tournament consensus is the mean of gN, gN-naive, RN, and QM2026, fixed before any revised agent was scored. The composition correction below was added after every agent had been scored; the supplement verifies that strategy ordering is stable across snapshotsreports all results without it.

Behavioral signals (stats-independent): *Hero diversity*: distinct heroes used, Shannon entropy of the pick distribution, and top-10 hero concentration across all drafts. *GD similarity*: agreement rate with the behavioral-cloning model at each pick step. These signals depend only on the strategy's own behavior and the fixed GD model.

A circularity deserves flagging: synergy and counter derive from the same aggregate pairwise statistics that populate the enriched features, so high measured synergy from an enriched-model agent partly reflects optimizing a supplied input. The behavioral signals diagnose collapse but cannot positively establish interaction-awareness. Two measurements break the circle: resilience, which scores each pick against the opponent's *future* picks, information no agent observes at decision time, and the head-to-head tournament of Section VII-A

a) Composition correction*:* A drafter that simply picks the highest win-rate heroes scored above every trained policy under a realized index, although 55% of its teams were structurally broken. We therefore checked every reference on the 1.94M snapshot games, which none of them trained on. Real teams with no healer, no frontline, or three heroes of a stacking role (2.0% of teams) win 4.9pp less often than RN predicts (±0.2), 5.5pp less than gN-naive, and 5.2pp less than QM2026; gN, which sees role counts, misses 2.2pp, and the consensus 4.4pp. RN's composition term shrinks each role multiset toward 50% with a 200-game prior, and degenerate multisets hold a median of 3–9 games. The penalty does not come from who fields such teams: matchmaking gives them nearly equally skilled opponents, and controlling for causal per-player skill estimates and pick position moves it from −4.79 to −4.83pp. We correct each reference with three structure terms (no healer, no frontline, stacked role), fitted as a logistic adjustment to the reference's own held-out predictions on the post-snapshot games. On snapshot games the corrected references are calibrated on degenerate teams (consensus gap +0.2pp, each reference within 0.6pp) and on normal teams, with log loss equal or lower. Every number below uses the corrected references (supplement: audit, fits, and uncorrected values). The terms are learned from human degenerate teams, 93% of which have one defect, as do 94–100% of the MCTS agents' degenerate teams; a team nobody fields, such as five tanks, gets only the sum of its indicators.

E. Search Methods

Greedysearch**:** For for each candidate hero, complete the draft via Generic Draft with behavioral-cloning (GD) behavioral-cloning opponents trained on the same replay snapshot (289→256→128→90, snapshot (5 seed variants), evaluate seeds), score the terminal state with V_θ, select the highest-scoring candidate.

the WP model, and pick the best. GD behavioral cloning: Argmax of the GD model at each step; the control for pure human imitation. CQL: Argmax **GD:** the behavioral-cloning argmax. **CQL:** argmax of a conservatively learned Q-function(no rollout needed; Q directly estimates action values).

MCTS: . **MCTS:** an AlphaZero-style [12] policy-value network (Section VII), ; the benchmark of Table V runs 200 simulations per moveat inference.

a) Sanity tests: We evaluate all value functions on hand-crafted scenarios: absurd compositions(5 tanks, 5 healers), traps(high-WR heroes in bad compositions), normal comparisons, and symmetrytests; 21 tests initially, expanded to , and the tournament agents play the policy argmax with no search at decision time. All WP models are also checked on 28 (Section **??**hand-built sanity tests (absurd compositions, traps, symmetry).

a) Data availability: Raw replays cannot be redistributed under the Heroes Profile terms of service, but the corpus is reconstructible through . Our corpus is a quota-limited subset of their public API: ranked Storm League drafts on patch 2.55, December 2021 through May 2026, with the exact 's replay range, so the released repository includes the replay-id bounds, the pinned aggregate-statistics snapshot, and processed feature matrices in the released repositorylist, split and fold assignments, and the code that rebuilds every statistic from those replays.

TABLE II

CROSS-EVALUATION AND COMPOSITION QUALITY Greedy cross-evaluation (480 DRAFTS PER STRATEGY; drafter, leak-free 256→128 MODELSOF TABLE I). LEFT: AVERAGE mean WP ASSIGNED BY the COLUMN MODEL TO ROW STRATEGY'S DRAFTS (BOLD: SELF-EVALUATION); Ind.: mean of the four consensus references. RIGHT: COMPOSITION METRICS; NAIVE-VS-ENRICHED rates. Naive-vs-enriched HEALER DIFFERENCE 15.818.3PP ($p < 10^{-6}$, $\chi^2 p < 10^{-9}$).ALL MODELS SELECTED FRONTLINE HEROES >92%.

Drafter	Naive	H. Str.	Enr.	Ind.	Heal%	Rng%	Deg%
Naive	**.638**	.629	.615	.564	62.1	77.1	60.8
Hero Str.	.621	**.642**	.612	.564	75.6	79.6	50.0
Enriched	.618	.618	**.648**	.570	80.4	77.9	47.5

IV. THE COMPOSITION GAP

A. *Cross-Evaluation*

Table II shows diagonal dominance, each strategy scoring highest on its own evaluator: greedy search optimizes for

a) Cross-evaluation*:* Each greedy drafter scores highest under its own model (Table II): search optimizes the specific model's value landscape. The enriched drafter scores lowest on both simpler evaluators (0.614, 0.610), sacrificing 4.0gives up 2.0pp of naive-scored win probability for compositional quality only the enriched evaluator perceives; no single-model metric can detect this trade-off.

A. *Composition Quality*

All three greedy drafters WP for a composition profile only its own model rewards, and the independent references barely rank the three drafters apart (0.564–0.570). All three fail composition safety, and the failure shrinks monotonically . Healer rate rises with feature richness across healer, ranged, and degenerate rates (Table II). This controlled comparison uses the smaller 256(62% → 128 models of Table I; the 51276% → 256→128 architecture used from Section **??** onward shrinks but does not repair these failures. Even the enriched greedy drafter leaves half of its drafts degenerate , the residual that Sections **??**–VII address80%), and the degenerate rate falls from 60.8% to 50.0% at the hero-strength step and to 47.5% with enriched features.

A. *The Accuracy-Policy Disconnect*

A sweep across

a) Accuracy does not predict policy*:* A fractional factorial over ten feature groups (Table III; the submission trained all 4,100 model configurations (2^{10} subsets of 10 feature groups × 2 architectures × 2 learning rates096 subsets with leaky features) shows that at 1.95M replays accuracy has essentially accuracy has stopped discriminating between feature sets: the 128 models span 57.1–57.7% on test, and no group's marginal impact exceeds 0.2pp; pairwise counters are largest (+0.20%) and per-hero win rate is *negative* (−0.16%). Yet the main effect exceeds 0.13pp. Role counts, which read no statistics, have the largest effect. The four pairwise groups each lose 0.12–0.22pp against the submission, which is where self-inclusion was concentrated. The policy consequences of the same features are massive: the enriched feature

TABLE III
Main effect of each feature group on accuracy (with minus without, pp), from a 2^{10-3} resolution-V fractional factorial (128 models, leak-free statistics, 256→128, seed 42). Main effects are unaliased with each other and with two-factor interactions; SE 0.02pp (test), 0.01pp (post). Submitted: the full-factorial marginal reported with leaky statistics.

Group	Test	Post	Submitted
role counts	+0.11	+0.13	+0.16
pairwise counters	+0.04	+0.03	+0.20
pairwise synergies	+0.03	+0.02	+0.15
map type	+0.02	+0.01	+0.01
map delta	−0.00	+0.02	+0.09
synergy detail	−0.00	−0.05	+0.16
team average WR	−0.03	−0.04	−0.05
counter detail	−0.07	−0.08	+0.14
hero WR	−0.08	−0.11	−0.16
hero-map WR	−0.09	−0.10	−0.02

set drops the WP assigned to a 5-tank team from 0.365 (naive) to 0.155, synthetic augmentation to 0.062 (Section **??**), and greedy-search degenerate rates separate by 14.6pp (Table II).

Among large. The leak-free models rate five tanks against a standard team at 0.28 (naive and enriched alike). Naive and enriched greedy policies agree on only 13% of bans and 7–13% of picks at matched states. Of the 50 drafts with the largest naive-vs-enriched evaluation gap, 64% had the naive model rates most above the enriched model, 76% have no healer and 86% paired individually strong heroes with degenerateteam structure: the naive model cannot represent the catastrophic interaction between "no healer" and "fifth damage dealer. " are degenerate.

A. *Independent Baseline*

a) Independent baseline*:* To verify generality, we retrain an independently designed value function: a We evaluate the HotS win-rate estimator released by Gourdeau,[2] predating the published discriminator (Section VI), a Siamese network (994K parameters) with multiplicative map conditioning over multi-hot inputs, originally reported at roughly about 61% accuracy on contemporaneous ranked-ladder data. Retrained on our snapshot it reaches 56.2% accuracy, within 1.6pp of on contemporaneous data, trained on our split and selected on validation. It reaches 56.5% on our test set (55.7% post-snapshot), 1.2pp below our naive model; the gap from the originally reported figure plausibly reflects data era and population, patch heterogeneity, and our stricter replay-level splitting, not the design. The OOD pathology is unambiguous: it rates a 5-tank team at 45.5%WP. It rates five tanks at 44.9%, nearly competitive(enriched model: 15.5%), and an earlier reproduction trained on 275K replays rated the same team at 52.8%, i.e., *favored*: more data attenuates but does not repair the pathology. In greedy evaluation , and under greedy search fields 40% degenerate teams (Table IV), the estimator shows partial interaction-awareness (positive synergy, near-zero counter) but fails composition safety with a 41% degenerate rate. The pathology is a property of the representation, not of any single implementationbelongs to identity-only inputs.

V. Closing the Gap: Features and Synthetic Data Why Value Search Builds Broken Teams

A. *Residual Gap Analysis*

[2]https://github.com/HairyBlob/HotS-Drafter

To provide a controlled baseline, we scale to a With a 512→256→128 architecture (311K parameters). Even with this capacitynetwork, the enriched model produces 22.0% degenerate compositions in greedy evaluation (Table **??**, no augmentation row), down from 49.6% at 256→128 but still substantial. Three factors contribute: search horizon, training concentration(yields 20.5% degenerate teams under greedy search with GD-argmax rollouts, and it rates five tanks against a standard team at 0.26. The cause is training concentration. Over 95% + of training teams include a healer), and feature granularity. Synthetic augmentation targets the second: the model never observes the `comp_wr` feature's pessimistic default correlated with actual losses, so it underweights the signal.

A. *Synthetic Augmentation*

Of the 252 possible 5-role compositions , the ∼100 per tier absent from the aggregates (Section III) are structurally degenerate and never drafted. We generate synthetic training records for each unseen composition: heroes sampled by role, paired against real opponent teams have a healer, outcomes assigned at a fixed win rate. Roughly 34K synthetic records (under 2% of training data) are added; the test set remains pure.

a) Scope: unseen vs. sparse.: A 27-configuration sweep over augmentation win rate, volume, and scope (supplement) shows that augmenting *unseen* compositions is the active ingredient; reinforcing sparse-but-observed ones leaves the 5-tank pathology unrepaired. The OOD problem in draft valuation is binary: compositions that appear at all in human play provide enough signal; compositions that never appear provide none.

a) Win rate landscape.: We then swept the assigned win rate over 0%, 5%, 10%, and and 118–167 of the 252 role compositions per tier have fewer than 50 % at the best scope (unseen) and volume (100), with the 512→256→128 architecture and games in our training split, so the composition features fall back to their pessimistic default on exactly the teams search should avoid, and the model has almost never seen that default paired with a loss. More features or capacity cannot supply outcomes that the 28-test sanity suite (Table **??**). Some unseen compositions may be unconventional-but-viable rather than structurally bad; the assigned win rate forecloses that possibility by fiat, a deliberate trade-off favoring safety.

A. *Augmentation Results*

Effect of synthetic augmentation across assigned win rates (5 seeds × 1000 drafts = 5000 per config, 512→256→128 architecture, unseen-only scope, volume=100, 28 sanity tests).

The extremes of Table **??** expose a calibration trade-off: WR=0% produces the strongest penalties (5-tank WP drops to 0.3%) but over-penalization distorts the value landscape, costing the most sanity tests (19/28) while leaving the highest augmented degenerate rate (18.3%); WR=50% minimizes degenerate rate (14.0%) but sits too close to the model's prior to penalize extreme compositions (5-tank WP 47.9%). We use WR=10% for the augmented models downstream: the best healer/degenerate balance with a still-strong penalty (5-tank WP 8.9%).

In greedy evaluation (Table IV), the augmented model matches the unaugmented enriched model's counter, synergy, corpus does not contain. The invalid region is small and easy to state, however: a team needs a healer, a frontline hero, and composition safety at a higher healer rate: augmentation does not sacrifice greedy-search interaction-awareness. Feature-ranged damage, and data-level pessimism together reduce the degenerate rate from 64% (naive)to 16% with no algorithmic changeat most two heroes of a stacking role. We therefore enforce it during search with a structural mask (Section VII-B). The mask removes every degenerate team, leaves the value function untouched, lowers measured synergy only slightly (greedy +0.36 →

+0.32), and wins head to head against the unmasked agent (0.511).

VI. Pessimistic Offline RL Collapses

The repair above required domain knowledge: engineered features and synthetic data constructed from role statistics. The offline RL literature suggests this should be unnecessary, since algorithmic pessimism is designed precisely to suppress out-of-distribution value overestimation from identity features alone. Our thesis is sharper: in terminal-reward assembly problems, algorithmic pessimism degenerates to behavioral anchoring by construction (Section **??**); this section quantifies the consequences: how tightly the anchor binds, that it tightens as the corpus grows, and that MCQ's penalty does not transfer at all.

A. *CQL Setup*

We reformulate drafting as offline RL. Each replay provides

a) Setup*:* Each replay gives 16 state-action-reward transitions; with team-swap augmentation this yields ∼62M training transitionstransitions, doubled with team swaps. The Q-network (311K parameters, same architecture as the controlled WP baseline) maps state (289 dims) through 324K parameters, 512→256→128 layers hidden layers) maps the 289-dim state to 90 Q-values , with invalid actions maskedto $-\infty$.

We use Monte Carlo returns with $\gamma = 1.0$: since reward is only at the terminal step and episodes are . With terminal-only reward and fixed-length , episodes, Monte Carlo returns ($\gamma = 1$) make the Q-target is simply the game outcome, making the objective exactly so CQL is supervised outcome regression plus a behavior-distribution penalty:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(Q(s,a), y) + \alpha \left(\mathbb{E}_{\mu}[\log \sum_{a} e^{Q(s,a)}] - \mathbb{E}_{\mathcal{D}}[Q(s,a)] \right)$$

behavior penalty:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(\mathrm{Q}(\mathrm{s},\mathrm{a}), \mathrm{y}) + \alpha\big(\mathbb{E}_{\mu}[\log \textstyle\sum_{\mathrm{a}} \mathrm{e}^{\mathrm{Q}(\mathrm{s},\mathrm{a})}] - \mathbb{E}_{\mathcal{D}}[\mathrm{Q}(\mathrm{s},\mathrm{a})]\big) \tag{1}$$

where y is the game outcome, α controls conservatism, and the second term penalizes high Q-values for actions not in the data. We sweep $\alpha \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$. The penalty 's only possible contribution is therefore anchoring The penalty can only anchor toward the behavior distribution, so landing near behavioral cloning is foreordained in kind; what the experiments quantify is how tight: GD agreement rose from 59–65% to 69–74% as the corpus quadrupled (Table IV).

b) Algorithm selection.: CQL, MCQ, and ; the experiments measure how tightly. We sweep $\alpha \in \{0.1, 0.5, 1, 2, 5\}$, MCQ's threshold $\tau \in \{0.4, \ldots, 0.8\}$, BC-regularized CQLare the natural discrete-action candidates: each operates directly on a Q-function over the finite action set with no actor network, and BC-CQL is effectively the 's $\beta \in \{0.1, 0.5, 1, 2\}$ (a discrete analogue of TD3+BC's behavioral-cloning regularization [15]. We also run IQL [10], the family designed to avoid over-conservatism, adapted to the same data and architecture ([15]), and a six-cell $\tau \times \beta$ grid). All six cells are statistically identical, and the mechanism is transparent: with terminal-only Monte Carlo returns, advantages are ≈ 0 and the AWR weights $e^{\beta(Q-V)}$ sit at 0.998–0.999, collapsing the policy objective to *literal* behavioral cloning regardless of expectile or temperature. The best cell (Table IV) reproduces the CQL profile and loses a five-opponent tournament cross-check (0.388 consensus WP)IQL grid [10].

A. *The Misleading Result*

Sweeping α with naive multi-hot features (5 seeds × 1000 drafts per setting), CQL achieves 93.0

a) CQL looks safe*:* With naive features CQL reaches 97.5–99.9% healer selection and 2.2–13.4% degenerate rate across the full range, with the best safety at $\alpha = 2.0$ (99.8%healer, 2.2%degenerate), far better than the enriched+augmented model's greedy-search results (Table **??**) and within noise of the GD baseline. From hero .7% healer and 1.9–9.3% degenerate across α (best at $\alpha = 2$: 99.6%, 1.9%), safer than any greedy value-search agent. From identity inputs alone , CQL it appears to subsume the entire feature engineering and augmentation feature pipeline.

A. *Rich Evaluation Reveals the Truth*

Comprehensive evaluation across both axes of draft quality (5 seeds × 1000 drafts = 5000 per strategy). Counter and synergy are average normalized pairwise deltas (positive = favorable); raw deltas carry a team-strength confound — strength-adjusted validation in the supplement confirms both predict outcomes; both are snapshot-dependent (Section III-C). Ent. is pick-distribution Shannon entropy; GD Sim is agreement with the behavioral-cloning model. Aug WP is the augmented model's evaluation of each strategy's drafts, a self-evaluation in the sense of Section **??**; the collapse case does not rest on it. Enriched greedy rows use the 512→256→128 architecture (Section **??**); Table II used 256→128.

Table IV reveals the full picture. The collapse is visible first in the stats-independent behavioral signals: CQLand

a) Rich evaluation shows collapse*:* CQL, BC-CQLproduce concentrated pick distributions and high , and IQL agree with the behavioral-cloning agreement (68.6–73.7% GD similarity), while the enriched models use the full hero pool with under 5% GD agreement. Notably, the same variants sat at 59–65% GD similarity on a 473K-replay predecessor corpus: quadrupling the data *tightened* the collapse.

The interaction metrics corroborate this picture: all collapsed methods produce model on 68–73% of picks, use 71–75 heroes with concentrated pick distributions, and have negative synergy (−0.16 0.13 to −1.600.18), the enriched models strongly positive synergy (+0.76 to +0.85), a sign flip; counter responsiveness separates by magnitude rather than sign same profile as GD itself (Table IV). Greedy and MCTS agents agree with GD under 5% of the time, use the full pool or a chosen head of it, and have positive synergy (+0.21 0.36 to +0.25 vs. +0.01 to +0.07).

A. *CQL with Enriched Features*

0.86). Adding the 86 enriched features to CQL (375 total dims) improves composition safety (2.3–2.5% degen)but does not produce interaction-awareness: synergy remains negative (−0.16 to −0.20), hero diversity stays concentrated (entropy 4.3–4.5, top-10 concentration 72–76%), and GD similarity does not budge (68.6–73.7%vs. 70.4–72.3% for naive CQL): enriched features sharpen CQL's picture of the behavior distribution's boundaries, and the pessimism penalty keeps the agent *inside* them.

The GD baseline calibrates both axes: its 99.3% healer rate confirms that composition changes nothing (3.4–3.6% degenerate either way); across three network sizes and five target-update rates the collapse signals hold (GD agreement 68–74%; supplement). The IQL cells are statistically identical, with advantage weights at 0.995–0.999; since the IQL Q-network's held-out loss (0.6948) is above that of a constant ($\ln 2$), the near-zero advantages may reflect an unfit Q-function as well as the absence of intermediate reward. GD itself shows that safety was never the hard problem, while its negative synergy(−0.20) shows that human drafters select by popularity and comfort rather than pairwise fit: 99.4% healer with negative synergy.

TABLE IV
Rich evaluation (5 opponent-sampling seeds × 1000 drafts per strategy). Counter and synergy use the community aggregates for every row. Ent.: pick entropy (bits); T10: top-10 hero share; GD: agreement with the behavioral-cloning model. Every row is trained on our split and selected on validation games. The anchored rows read no aggregate statistics except CQL (enr.), trained on out-of-fold features; greedy and MCTS rows use leak-free value functions. MCTS rows: the five F seeds (Section VII), policy argmax.

Strategy	Heal%	Deg.%	Counter	Synergy	Dist.	Ent.	T10%	GD%
GD baseline	99.4	1.3	−.02	−.16	66	4.50	71.7	81.4
G&A discriminator	99.9	0.3	−.02	−.06	87	4.39	75.3	40.6
CQL α=.5 (naive)	99.7	3.6	−.01	−.17	75	4.62	69.1	69.6
CQL α=1 (naive)	99.6	3.4	−.01	−.17	71	4.59	70.2	70.8
CQL α=.5 (enr.)	99.9	3.4	−.02	−.18	71	4.65	67.0	68.4
CQL α=2 (enr.)	99.0	3.6	−.01	−.18	71	4.67	66.9	70.4
BC-CQL β=1	99.0	1.5	−.01	−.16	71	4.62	68.5	73.2
IQL τ=.9 β=3	99.8	3.1	+.00	−.13	75	4.67	67.9	70.6
MCQ τ=.5	0.0	100	−.05	−.88	10	2.73	100	0.0
Gourdeau estimator (greedy)	66.8	40.1	−.08	+.36	90	5.73	43.2	4.9
Enriched (greedy)	87.2	18.6	+.09	+.36	90	6.01	33.7	4.8
MCTS	96.8	3.8	+.06	+.86	35	3.68	85.6	3.4
Constrained MCTS	100	0.0	+.04	+.71	40	3.70	86.0	4.0

A. *MCQ: A Different Collapse*

MCQ [9] penalizes only Q-values exceeding a threshold τ, replacing CQL's log-sum-exp penalty with $\sum_a \mathrm{ReLU}(Q(s,a)-\tau)$. We sweep $\tau \in \{0.4, 0.5, 0.6, 0.7, 0.8\}$, the full attainable range of sigmoid-bounded Q-values, with the same architecture and data as CQL, the natural discrete instantiation of its pseudo-target construction.

MCQ's penalty structure

a) MCQ collapses differently*:* MCQ's penalty does not transfer to this discrete, sigmoid-bounded setting: two distinct failure regimes emerge depending on τ. Thresholds discrete Q-values. For $\tau \geq 0.6$ provide zero effective penalty, reducing the loss to pure Bellman regression and yielding it is inactive and yields a narrow policy over the same ∼9 heroesregardless of context8 heroes; at $\tau = 0.6$, 126 of the 128 units of the last hidden layer are zero on 200,000 held-out states. At $\tau = 0.5$ the penalty is active but destructive: it drives Q-values toward zero, and the near-zero pre-activations leave most late-layer neurons permanently deadit drives Q toward zero: all 128 units of the third hidden layer and 255 of 256 of the second are zero on those states, so the Q-values reduce to the output bias, a fixed hero ranking. The resulting policy (Table IV) is near-deterministic over uses 10 distinct heroes: 100% healer selection yet 45.4% degenerate through heavy role stacking, with the worst synergy we measure.

We also evaluate BC-CQL, which adds a behavioral-cloning loss to the CQL objective: $\mathcal{L}_{\mathrm{BC}} = \mathrm{NLL}(\mathrm{softmax}(Q), a_{\mathrm{data}})$, sweeping $\beta \in \{0.1, 0.5, 1.0, 2.0\}$: near-perfect composition safety, high GD agreement, and negative synergy (−0.16, Table IV); even explicit behavioral-cloning regularization cannot produce interaction-awarenessheroes, never takes a healer, and fields only degenerate teams.

A. *The Professional-Manifold Discriminator*

a) The professional-manifold discriminator*:* We implement Gourdeau and Archambault's published draftertakes a third route to anchoring: a discriminator trained to distinguish real drafts from randomly perturbed ones, drafting by minimizing the generated-probability of the resulting statenon-GAN drafter [5]. We implement their stronger non-GAN variant on our snapshot. It is the safest learned strategy we measure, 100% healerand 0.0% degenerateacross 5(99.9% healer, 000 drafts (Table IV) , but its 0.3% degenerate) and shows no interaction-awareness is null (synergy +0.09), and training it longer tightens the anchor (hero diversity 80 → 53, GD agreement 44% → 51%). Imitation, algorithmic pessimism, and manifold discrimination are three distinct mechanisms that land in the same place: (synergy −0.06). Each of these methods reaches safety by anchoring to human behavior, without compositional reasoning.

A. *Robustness of CQL Collapse*

CQLcollapse is not an artifact of a specific architecture: sweeping three network capacities crossed with five target-network update rates at $\alpha = 1.0$ (15 configurations), the behavioral collapse signals are stable throughout (entropy 4.25–4.52, GD agreement 69.0–77.3%), no configuration produces interaction-aware behavior, and the update rate is inert, consistent with Monte Carlo returns (no bootstrapping signal)observed play. CQL's penalty treats every uncommon action alike: it cannot tell a fifth tank from a niche counter-pick whose matchup is rarely seen.

Three offline-RL families, explicit pessimism (CQL), thresholded pessimism (MCQ), and expectile-based value learning (IQL), thus converge on the same behavior-cloned policy, each by its own mechanism.

VII. MCTS WITH THE REPAIRED Leak-Free VALUE FUNCTION

A. *MCTS Training*

We train an AlphaZero-style

a) Training*:* The network (4.4M parameters) via self-play with MCTS. The network shares a residual backbone (289 dims → 768, 3 residual blocks, compressed to blocks, → 256) between a perspective-invariant policy head (256 → 90) and a perspective-aware value head(257 → 128 → 64 → 1, receiving the backbone output plus an our-team indicator). During . In self-play , the network plays it drafts one side with MCTS while the opponent samples from against a pool of 5 GD variants. Training against behavioral-cloning opponentsrather than pure self-play is deliberate: the deployed assistant faces human opponents, and opponents. BC opponents keep trajectories within near the WP model's training distribution ; the cost, exploitability of a static opponent, is mitigated by sampling the seed-diverse pool per episode and quantified by the tournament of Section VII-A. Configurations labeled enriched_full use the enriched WP model*without* synthetic augmentation; H_augmented adds it, isolating augmentation's effect under deep search.

We train 13 configurations varying the WP model, simulation budget, training length, network capacity, policy head depth, and enriched-feature scope, each evaluated across 15

MCTS experiment results (15 seeds × 600 drafts per configuration, all tiers randomized, 200 sims at inference). WP model = value function used during training; Avg WP = mean ± across-seed std; Resil. = early-pick resilience (Section III-C); Dist. = distinct heroes per seed. C_large doubles the backbone width; D_deep uses a deeper policy head.

TABLE V

MCTS benchmark (5 seeds, E and in-sample rows 3 seeds, × 1,000 drafts; 200 simulations at inference, root temperature 1). Train: the value function the configuration optimized (K: the hero-identity model). gN, RN, QM26: composition-corrected references (Section III-D); gN seed SE 0.002–0.003.

Config	Sims	Train	gN	RN	QM26	Deg%	Syn.
Leak-free value function							
B	200	.723	.665	.641	.664	10.5	+1.06
F	400	.740	.662	.641	.669	4.5	+1.29
J	800	.751	.665	.642	.668	7.9	+1.31
E (1M ep.)	200	.726	.659	.634	.656	9.2	+1.21
K (no feat.)	400	.726	.660	.643	.672	15.4	+1.24
In-sample statistics (leak control)							
B	200	.742	.656	.633	.648	11.4	+1.07
F	400	.785	.653	.631	.665	11.7	+1.56
J	800	.811	.636	.619	.671	27.1	+2.02

random seeds (600 drafts per seed against behavioral-cloning opponents, and match the deployed setting, at the cost of exploitability by an adaptive opponent. WP evaluation and value head are symmetrized across team order. Every configuration is trained on our split against the leak-free enriched model, with its statistics in the kernel's lookup tables: B, F, and J run 200, 400, and 800 simulations per move for 300K episodes (5 seeds each), and E runs 200 simulations per move at inference). Both the WP evaluation and policy value head are symmetrized across team perspectives to enforce $P(t_0, t_1) + P(t_1, t_0) = 1.0$, removing team-order bias (up to 24% asymmetry otherwise)for 1M episodes (3 seeds). Two configurations isolate single factors: K matches F but searches against the hero-identity model, which reads no statistics (5 seeds), and B, F, and J are repeated against the enriched model with in-sample statistics (3 seeds each), the leak of Section III-B at paper scale.

A. *MCTS Training Results*

a) Benchmark*:* Each checkpoint drafts one side of the same 1,000 configurations (mid tier, random map and side) against the GD pool with 200 simulations per move and its own value function at the leaves. We benchmark every seed; no seed is selected. Within a run, the training worker keeps the checkpoint with the best periodic evaluation under the run's own value function, the submission's rule, which favors checkpoints that score well under that value function. Every draft is scored by the independent references. "Train" is the value function the configuration optimized: the submission's only reported metric.

Table V shows results across 11 of the 13 configurations (195 total runs). The WR column, like Avg WP, is adjudicated against the static behavioral-cloning opponent pool ; its saturation near 99.9% is exploitability-limited, a training diagnostic.

b) Simulation scaling.: The dominant finding is that more MCTS simulations during training monotonically improves quality. At 300K episodes with the enriched_full value function , increasing from 200 to 800 simulations per move yields 0.726→0.753→0.764→0.768 average WP (+0.042 total, $t = 11.2$, $p < 0.001$ across 15 seeds), with diminishing returns per step. Synergy exploitation and early-pick resilience scale in parallel (0.028 ± 0.005). A judge is a noisier model than that value function, so even an honest gain shrinks under it: on the test replays gN moves 0.59 per unit of the leak-free value function (consensus 0.55; supplement), which predicts about +1.20→0.016 for this step. Measured, the independent references stay flat: from 200 to 800 simulations gN changes by +1.44 synergy, 0.000 (±0.004), RN by +0.82→0.000 (±0.006), and the consensus by +0.95 resilience from 0.003 (±0.004). The 200 to → 400 step lowers the degenerate rate from 10.5% to 4.5%, and the 400 → 800 sims), and counter responsiveness rises from step raises it to 7.9%. Synergy, which reads statistics that overlap the value function's, keeps rising.

The leak control shows what in-sample statistics do to the same pipeline. At matched simulations the leaky value function raises the agents' own score (+0.03 to 0.020, +0.07: deeper search selects early picks the opponent cannot profitably answer.

c) Training length.: Longer training also helps but is the weaker axis: 0.045, and +0.060 at 200sims, extending from 300K to 1M episodes improves WP from 0.726 to 0.752 (+0.025), roughly what doubling the simulation budget buys, for 3.3× the episode cost.

c) Capacity and policy head.: Doubling the backbone width (C_large, 0.728, 400, and 800 simulations) and lowers gN and RN (at 800: gN −0.029 ± .011)is statistically indistinguishable from the base network (0.7260.006, RN −0.023 ± .013)at the same episode budget: larger capacity is neutral at fixed search depth. A deeper policy head (D_deep, 0.664) remains clearly harmful, achieving excellent safety (5.3% degenerate) but the lowest interaction-awareness of any enriched configuration (synergy +0.40, resilience +0.21), locking onto the safe modal composition rather than exploiting matchup structure. At fixed compute, simulation budget, not network size, is the binding constraint.

c) Value function quality.: The enriched_full value function (B_fullwp, 0.7260.006, consensus −0.015 ± .013 WP) vs. the true base model with identical architecture but no enriched features (K_truebase, 0.703±.012 WP) shows a +0.023 WP gap ($t = 5.0$, $p < 0.001$; 15 seeds each0.006). The gap is real but far smaller than the +0.070 we measured on a 473K-replay predecessor corpus, where the true base agent collapsed to 26% healer and 75% degenerate compositions. Data scale substantially closes the value-quality gap under deep search but not the OOD gap: at 1.95M replays the true base agent still produces 20% degenerate compositions (81% healer) against 8–12% for the enriched configurations, and features remain decisive for shallow search (Section **??**). To decompose the gap within the feature set, M2_relational and N2_absolute retrain the value function at identical settings on complementary halves of the enriched features (relational: pairwise counters/synergies, counter detail; absolute: team-average, meta, composition win rates; role counts in both arms). Neither family alone recovers the full model (0.719widens with search. From 200 to 800 simulations the leaky agents' own score rises by 0.068 while gN falls by 0.020 (±.024 0.006) and RN by 0.015 (±0.007), and 0.714at 800 simulations 27% of their teams are degenerate. QM2026 alone does not penalize them. The leak therefore inflated the self-reported gains, and with it more search over-optimizes. With leak-free statistics more training simulations neither help nor hurt the independent references at this scale. A value function

calibrated on human games is not thereby calibrated on the drafts search produces.

c) Training length and root selection*:* Training for 1M episodes instead of 300K at 200 simulations does not help: E − B is −0.005 (±.008 0.003) under the consensus (gN −0.006, RN −0.007), with a similar degenerate rate (9.2% vs. 0.726), and the gap over K_truebase decomposes roughly additively: 10.5%). Taking the argmax of root visit counts at inference adds 0.000–0.004 gN over sampling at temperature 1 (mean +0.016 relational ($t = 2.3$, $p = 0.04$) and 0.002) and never lowers gN. Our operating point is F with argmax root (the tournament plays the policy argmax without search): it ties J under every reference at half the training compute and has the lowest degenerate rate (4.0% with argmax root).

d) Features*:* The submission reported that enriched features add +0.010 absolute ($t = 2.8$, $p = 0.01$) approximately sum to the full 0.023 WP over an identical featureless agent, judged by the enriched agent's own value function. At matched search the leak-free features add nothing under the independent references: F − K is +0.023. Only N2's deficit against the full model reaches significance ($t = 3.3$, $p = 0.004$).

e) Augmented model.: The augmented WP model (H_augmented, 0.721 WP) achieves the best safety of any 200-sim enriched-family configuration (95% healer, 8.3% degenerate) but slightly lower overall WP than enriched_full at the same simulation budget: the synthetic augmentation that helps greedy search(Section **??**)is partially redundant for MCTS, which learns implicit composition regularization through self-play. 0.002 (±0.004) under gN, −0.002 (±0.006) under RN, −0.002 (±0.004) under QM2026, and −0.001 (±0.003) under the consensus. What the features buy is composition safety. The featureless agent fields 15.4% degenerate teams against 4.5% (−10.9 ± 3.3pp across seeds; healer rate +10.0 ± 3.1pp), because a value function that reads no role statistics has no way to see that a team lacks a healer.

e) Implementation.: These experiments are enabled by the fused CUDA implementation of contribution 4: 140–190 episodes/second per GPU depending on multi-GPU contention, 2.2× a tuned batched-evaluation host baseline doing identical per-simulation work at equal search quality; the controlled comparison and throughput progression appear in the supplementary material.

A. Round-Robin Tournament

The win probability results above are adjudicated by our own value functions against behavioral-cloning opponents, a circularity we address with a direct head-to-head tournament among eleven strategies: MCTS, enriched and enriched+augmented greedy, CQL (naive and enriched) , GD, MCQ, the two external reimplementations (Gourdeau's win-rate estimator under greedy search, and the G&A manifold discriminator) , and constraint-masked variants of MCTS and enriched greedy (Section VII-B) . Every ordered strategy pair plays

Eleven strategies play every ordered pair for 200 drafts (both first-pick assignments, stratified maps and tiers; 110 ordered pairs), terminal drafts scored by all four WP variants and the full metric suite of Section III-C, so no single model's judgment decides the outcome. pairs): MCTS and constrained MCTS, enriched and constrained greedy, the featureless MCTS agent K, the Gourdeau estimator and the G&A discriminator, CQL (naive and enriched), GD, and MCQ. Agents were fixed before any tournament score existed, and every agent is trained on our split. Both

TABLE VI
Tournament: mean win probability over each strategy's 20 ordered pairs, and pairs won under the consensus (mean of gN, gN-naive, RN, QM2026; SE 0.001, clustered by draft configuration, without seed-to-seed variation of the agents). All judges composition-corrected (Section III-D). Deg.: degenerate rate of the strategy's teams.

Strategy	Cons.	Won	gN	RN	QM26	QM21	Deg.	Syn.
Constr. MCTS	.612	20	.614	.598	.625	.682	0.0	+.75
MCTS, no feat.	.607	17	.593	.589	.639	.687	17.2	+.98
MCTS	.606	17	.595	.588	.634	.684	9.2	+.89
Constr. greedy	.541	14	.556	.545	.511	.484	0.0	+.49
Enr. greedy	.529	11	.540	.531	.504	.478	22.4	+.55
Gourdeau est.	.521	11	.523	.513	.512	.504	44.9	+.61
CQL naive	.459	8	.456	.458	.472	.466	3.9	−.15
GD	.444	4	.442	.448	.453	.442	4.1	−.22
CQL enr.	.444	4	.441	.448	.453	.432	6.0	−.22
G&A discrim.	.442	4	.441	.449	.447	.429	0.6	−.08
MCQ	.295	0	.301	.333	.251	.212	100	−.42

MCTS agents use the five F seeds, pooled (draft i uses seed $i \bmod 5$), the operating point of the benchmark; K pools its five seeds the same way. No contestant's value function is a judge.

Constrained MCTS wins all 20 of its ordered matchups at a consensus (mean over the four evaluators) WP of 0.670 ± 0.002 (SE across its 4, 000 tournament drafts), and the margin is stable under each evaluator individually: 0.657 naive, 0.666 hero-strength, 0.678 enriched, 0.677 augmented. The naive and hero-strength evaluators share *no* features with the underlying value function beyond hero identity: the lead is not self-graded. Consensus standings (across-draft SEs ≤ 0.003): constrained MCTS 0.670 (20/20 matchups won) >

Three tiers separate under every reference (Table VI): MCTS 0.658 (18/20, losing only the two orderings against its constrained variant) > constrained greedy 0.595 (16/20) ≈ enriched greedy 0.595 (14/20) > enriched+augmented 0.580 > G&A estimator 0.519 > CQL enriched 0.420 ≈ CQL naive 0.419 ≈ G&A discriminator 0.417 ≈ GD 0.404 > MCQ 0.225 (0/20). The two external baselines land by mechanism rather than authorship: the win-rate agents at 0.61, searching greedy agents at 0.52–0.54 (the Gourdeau estimator, the only one external baseline that searches a value function, finishes above 0.5 (10/20 ordered matchups won), ahead of every anchored method; the discriminator falls inside the among them), and every behavior-anchored cluster.

A final check uses judges from outside the corpus family entirely. Quick Match is a separate game mode with no draft phase: players queue with a chosen hero and the matchmaker assembles teams, so its games contain no ban-pick sequences, no draft-stage selection pressure, and no overlap with the replays any model above trained on. We train win-probability judges on this mode with deliberately naive features (hero multi-hot, map, skill tier; no aggregate statistics), one on 91method at 0.44–0.46, with MCQ last. Rank correlation with the consensus over all eleven strategies is 0.94–0.99 for every reference, both Quick Match judges included, which share no game with any model here. Within the top six the Quick Match judges agree less (0.66): they place the greedy agents near or below 0.5 and K first. They confirm that searching a value function beats anchoring to human behavior; they leave the finer ordering among searchers open. MCTS and K tie head to head (0.500 ± 0.003; gN 0.501, RN 0.498, 365 games from 2021 and one on 116QM2026 0.492), 589 games from the current era, and score the tournament drafts with each. Both reproduce the

consensus ordering at Spearman 0.92, and both score the degenerate probes as losing (current-era judge: five mages 0.38, five tanks 0.41 against a standard composition). Their absolute scale compresses toward 0.5, as expected from a judge that has never seen a draft, but the ordering they induce is the consensus ordering, from a corpus the agent has never touchedas in the benchmark, while K fields nearly twice as many degenerate teams (17.2% vs. 9.2%). Before the composition correction K ranked first (consensus 0.602 vs. 0.597), because the uncorrected references barely charged its broken teams; constrained MCTS beats it head to head (0.511 $\pm$ 0.003).

B. Structural Constraint Masking

Gourdeau and Archambault 's discussion proposed constraining win-rate search to viable compositions [5]; we implement that proposal on top of the learned agent as a fourth repair layer, substituting structural validity, a weaker and cheaper constraint, for their professional manifold. At . We implement a structural version, which is our recommended repair: at each of our pick steps, a candidate picks, a hero is masked out iff picking it would leave no completion of the team from the remaining pool with at least one healer, one frontline, and one ranged-damage hero and at most two heroes per stackable role (reachability decided exactly by memoized enumeration). This is validity masking, not behavioral anchoring: it The mask prunes only the structurally invalid regioninvalid region, by rule, leaving the policy free everywhere else; interaction-awareness survives where anchoring, which prunes toward human behavior, loses it. Safety is thereby a guarantee, not an empirical rate: and guarantees 100% healer and 0.0% degenerate hold by construction. The empirical findings are two. First, interaction-awareness survives the mask (0% degenerate teams. Interaction-awareness survives it: constrained MCTS keeps synergy +0.95, counter 0.71 (+0.20; masked greedy : synergy 0.86 unmasked) and the same pick entropy (3.70 vs. 3.68 bits), and masked greedy keeps +0.74 over 90 distinct heroes), where behavioral anchoring destroys it. Second, the constrained agent beats its unconstrained counterpart head-to-head: constrained MCTS wins all 20 tournament matchups, including both orderings 0.32, while the anchored methods of Table IV sit at -0.06 to -0.18 (MCQ -0.88). The mask costs no win probability and gains a little: head to head with the same checkpoints, constrained MCTS scores 0.513 $\pm$ 0.004 against unconstrained MCTS(0.577 and 0.542 consensus WP); constrained greedy wins 16/20 vs. 14/20 unconstrained. The cost is pick diversity: entropy 3.84 over 32 distinct heroes (top-10 concentration 84.5%), and masked greedy 0.511 $\pm$ 0.003 against unmasked greedy. Under the uncorrected references both pairs tied (0.497, 0.500), because those references barely charged the broken teams the mask removes.

VIII. Discussion Expert Judgment

A. Two Separable Axes

Every judge in this paper is a model. We tested whether experienced players could serve as a check that is not, in a study registered before data collection (https://osf.io/fspuc). The agents were the ten tournament strategies without K, retrained with the same recipe on the ranked games before 2026-09-01 so that the drafts reflect the meta raters play (patch 2.55.17); the MCTS agent is F at 400 simulations. The 901 items were 280 pairs of drafts from a tournament among these agents, 570 real

TABLE VII
Expert study (9 raters). Top: agreement, the share of judgments on model-generated pairs that pick the draft the model label prefers. Bottom: the share that pick the winner of a real game; the model column is the label's accuracy on the same 190 games per tier (descriptive, post hoc; Wilson 95% CI). Expert estimates are mixed-model estimates with 95% CIs; counts are items / judgments.

	Items / judg.	Experts	Model label
Agreement with the model on generated pairs			
All clear pairs	255 / 488	.50 [.43, .57]	–
Anchored vs. searching	113 / 215	.33 [.21, .48]	–
Picking the winner of real games			
Low tier	190 / 387	.60 [.54, .65]	.60 [.53, .67]
Mid tier	190 / 387	.49 [.43, .55]	.54 [.47, .61]
High tier	190 / 387	.53 [.46, .59]	.56 [.49, .63]

ranked games from 2026-09-14 to 09-27 with known winners (190 per tier), 40 shared calibration games, and 11 attention checks that pair a real draft with a role-broken team. Fourteen paid raters each had a fixed set of 240 items, so every machine pair was rated three times, and for each item picked the better draft, gave a win probability, and stated a confidence, without being told which drafts were real. A machine pair's model label is the mean of three draft-only value functions (hero identity, hero strength, enriched) trained out of fold on the same snapshot and composition-corrected as in Section III-D; pairs within 0.02 of 0.5 are near-ties and excluded.

Safety vs. interaction-awareness for each strategy. Marker size encodes hero diversity; shape and color distinguish method families. Only MCTS configurations reach both low degenerate rate and strong positive synergy; constraint masking (Section VII-B) pins the degenerate rate at zero.

Fig. 2 summarizes every method on both axes, which require fundamentally different solutions. Safety is trivially solved by anchoring to human behavior, and the tighter the anchor the safer (discriminator, then GD, then CQL/BC-CQL) , or made exact by structural validity masking (Section VII-B); MCQ's non-transferring penalty fails even at safety. Interaction-awareness requires value function search with domain-structured features; no anchored variant achieves it Eleven raters completed all 240 items. We closed data collection on 2026-10-26, about three weeks before the registered six-week limit, after the three remaining raters did not respond to a reminder; no outcome results were examined before closing. Nine raters passed the registered gates (at least 7 of 8 screening items, 2 of 3 late attention checks, and a median of 10 seconds per item); two failed the screener. The three who stopped early (7, 24, and 63 items) had not reached every gate and are excluded; adding the one who passed every gate they reached changes no result. Estimates come from mixed-effects logistic models with rater and item random effects (Table IVVII; analysis notes in the supplement).

A. Why Pessimistic Offline RL Collapses

CQL's pessimism penalizes *all* uncommon state-action pairs, not just bad ones: it cannot distinguish an action that is uncommon because it is structurally bad (a fifth tank) from onewhose matchup rarely arises (a niche counter-pick)*Can experts check the models?* Only if they can judge drafts. On the 570 real

games experts picked the winner 60% of the time at low tier (Holm-adjusted $p = 0.004$), 49% at mid tier, and 53% at high tier, 53% overall. A draft-only model cannot do much better: our value functions reach 57–58% on held-out games, and on these same games the model label scores 57% overall (60%, 54%, 56% by tier; the enriched value function alone 59%). With 190 games per tier those model figures carry intervals of about $\pm$7pp, so the tiers do not separate experts from models; above the lowest tier the experts are at chance. Their win probabilities did not track the model's (slope 0.00, 95% CI -0.22 to 0.22), and raters chosen in advance for low-rank experience were no more accurate at low tier (57% vs. 59%).

Agreement on generated pairs. Experts picked the model-preferred draft in 244 of 488 judgments (0.50, and it cannot substitute for domain-structured features that encode *why* compositions work95% CI 0.43–0.57), so the registered primary test fails. The result does not move with the near-tie threshold (0.49–0.52), with any single value function as the label (0.50 each), or without the seven heroes patch 2.57 changed (0.49). Where a behavior-anchored strategy (GD, either CQL, or the discriminator) met a searching one, experts chose the anchored draft in 152 of 215 judgments (agreement 0.33, 0.21–0.48, $p = 0.03$); on the other pairs they agreed with the model 57–66% of the time (exploratory). A Bradley–Terry ranking of the ten strategies from their choices runs against the tournament order (Spearman $\rho = -0.19$). Agreement did not fall on items the value-function ensemble found unfamiliar (slope $+0.24$ per SD of ensemble variance, CI -0.02 to 0.50), which meets the registered robustness criterion; with agreement at chance it shows only that the disagreement is the same everywhere.

A. *What Transfers Beyond HotS*

What it means. The experts consistently preferred human-like drafts, but they could not pick the winners of real games above the lowest tier, so their preference is no evidence about which drafts win. The check is uninformative: it neither confirms nor refutes the models' ordering. As registered, the evaluation claims of this paper rest on outcome-based evidence: held-out accuracy and calibration, the composition audit on real games, and the independent references, which share no data with any agent but are themselves models.

Three findings are properties of the problem structure, not the game: the safety/interaction-awareness decomposition, since any

IX. Discussion

a) Two axes*:* Anchoring to human behavior provides safety, and a rule-based mask makes it exact (Fig. 2). Interaction-awareness requires searching a value function; no anchored variant reaches it. Some findings depend only on the problem structure. Any assembly domain separates structural validity from responsiveness to the opposing set; the anchoring-collapse mechanism, since any . Any terminal-reward assembly problem trained with Monte Carlo returns

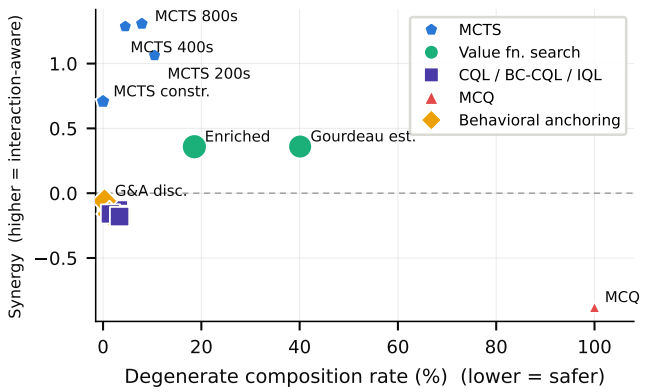


Fig. 2. Safety vs. interaction-awareness (Table IV; MCTS 200s/400s/800s are the leak-free configurations of Table V, searched at inference with root temperature 1, while constrained MCTS is the policy argmax of Table IV; anchored points are the retrained baselines of Table IV). Marker size encodes hero diversity. Only MCTS reaches both low degenerate rate and positive synergy; masking pins the degenerate rate at zero.

leaves pessimism only the anchoring pathway; and the validity-masking principle, applicable . Validity masking applies wherever the invalid region is rule-describablecan be described by rule. The interaction metrics also port , requiring port too, needing only pairwise outcome statistics and a reference behavior policy. The domain content does not transfer: the role taxonomy, specific the features, and aggregate the composition heuristics must be rebuilt per domain, and a domain without a large second source of aggregate statistics loses the never-observed-composition signal that targets synthetic data.

A. *Practical Implications*

For offline RL practitioners, low degenerate rates may indicate behavioral cloningrather than learned reasoning; reporting action diversity alongside rewarddetects this collapse.

The collapse

a) Practical lessons*:* Low degenerate rates in offline RL can mean behavioral cloning; report action diversity next to reward. Anchoring is a limitation only when the behavior policy is suboptimal, as here , where anchoring inherits the here because π_β reflects comfort-driven habits of uncoordinated queue playqueue habits; where the behavior policy is near-optimal(medical protocols, industrial control), , staying near it is the right objective.

Contribution 4's deployment target is a free goal. Anyone building a value function from aggregate outcome statistics and then searching it should compute the statistics out of fold, check calibration on games outside them, and judge the policy with references the search never saw. In our submitted pipeline the leak cost half a point of accuracy, was invisible on a leaky test set, and inflated the value function's own measure of every gain; independent references confirm neither the feature gain nor the simulation gain, and with the leak more search makes the agents worse. Where invalid compositions can be described by rule, mask them during search; the training data cannot teach a value function about states it never contains. The

deployed web tool[3] running runs MCTS in-browser at 50–200 simulations per move.

A. Open Problems

The talent stage. Talent selection at seven Open problems include the in-match levels refines the draft-stage assembly our corpus records. *Patch-driven non-stationarity.* Balance patches shift the reward landscape while preserving combinatorial structure, a natural continual-learning testbed. talent stage and patch-driven non-stationarity.

X. Conclusion

Adversarial team assembly poses a distinctive challenge for learned policies: an exponential state space , data clustered as well as sparse, and a terminal evaluation that In team assembly the state space is huge and the terminal evaluation must itself be learned from clustered data. In MOBA drafting, search actively exploits the out-of-distribution optimism of learned value functions to assemble structurally broken teams, and the textbook remedy, algorithmic pessimism , algorithmic pessimism does not repair the failurethis: with Monte Carlo returnsthe CQL penalty can only anchor toward , CQL anchors to the behavior distribution, MCQ collapses to a near-deterministic ten-hero policy, and IQL 's vanishing advantages reduce it IQL reduces to behavioral cloning.

The repair that works is layered: domain-structured features give the value function the vocabulary to represent compositional quality; synthetic data supplies safety for shallow greedy deployment; , and MCQ collapses to ten heroes. What works is search plus a structural mask: MCTS self-play converts the value function into a policy that is safe and interaction-aware at once; and structural constraint masking removes the residual invalid picks by rule, without anchoring to behavior. The headline evidence is the tournament: with the constraint mask , the agent is structurally valid by construction, preserves interaction-awareness, and sweeps all twenty head-to-head matchups at 0.670 consensus win probability. The enriched features contribute over a value function with domain features finds interactions, and the mask removes the broken teams search still builds. Judged only by references no agent trained against, MCTS agents win 17–20 of 20 ordered matchups at 0.61 win probability, far above every anchored method (0.44–0.46), and the mask makes them valid by construction with a small gain. A registered expert study could not check that verdict: the raters agreed with our value functions at chance, but they also picked real winners at chance above the lowest tier, so the claims rest on outcome-based evidence. The first version of our pipeline built features from statistics containing each game's own outcome. Rebuilt that way, its own value function credits 800 over 200 training simulations with +0.023 WP even under deep search ($p < 0.001$), a gap that data scale narrows but does not close. 0.068 win probability, while gN measures -0.020 and 27% of the 800-simulation agent's teams are degenerate. With out-of-fold statistics the own score still rises by 0.028 and the independent judges stay flat. Domain features in the value function leave independent win probability unchanged at matched search and cut degenerate teams by about two thirds. Value functions learned from aggregate outcomes and then searched should be built on out-of-fold statistics and checked for calibration; their agents should be scored by independent judges, and the simulation budget chosen against those judges. The judges need the same audit: ours under-charged real broken teams by about 5pp until we measured them on such teams.

Limitations. Single gamedomain. All adjudication, including the tournament, uses value functions trained on our own corpus and behavioral-cloning opponents; two of the four evaluators share no features with the winning agent, but all four share its training corpus, so bias-sharing without feature-sharing remains possible. Two independent checks bound this concern. The Quick Match judges (Section VII-A)reproduce the tournament ordering from a disjoint corpus in a mode with no draft phase. And a 20-model value-function ensemble varying seed, architecture, feature subset, and training-data half (supplement) bounds it from within the corpus : the agent's drafts draw near-human ensemble predictive variance, degenerate compositions are flagged at 3.2× the human medianOne game. The independent judges are weaker than an in-corpus model (54.0–55.7% vs. 57.6% held-out) and each has a blind spot (Section III-D); their agreement is strong for the coarse ordering and weaker among searching agents. Their composition terms are fitted on human teams with one missing role, so they extrapolate to rarer structures, and the fitted penalty is about 2pp too large on games after the 2.55.17 balance patch. Post-snapshot games differ from the corpus in population, though not in patch. Tournament agents play the MCTS policy without search, and all eleven tournament strategies sit well below the degenerate regime. The agent's hero-pool concentration reflects ground-truth-validated synergy edges (supplement); only its exact five-hero compositions are unobserved in human play. Our validation proceeds in three tiers: statistical validation of the metrics and the ground-truth synergy analysis,already in hand agents trained against behavioral-cloning opponents, so exploitability by adaptive opponents is untested. The leak-free MCTS runs have three to five seeds (SE resolves about 0.01). Earlier in-sample checks of the agent's favorite hero combinations did not replicate on later games and are withdrawn. On 11,034 drafts from an amateur league with persistent five-player teams (supplement); a blinded expert study in progress, designed with known-outcome anchor drafts because professional judgment is itself tier-limited; and longitudinal forward prediction under real use, pursued through an amateur league (, whose games are in neither our corpus nor the aggregates, the Nexus Gaming Series)but requiring lead time. An external evaluation on 11,034 coordinated-league drafts (supplement)shows modest degradation under distribution shift (58.1% → 55.5%), concentrated in the least-ladder-like drafts. The composition-quality heuristics depend on community-aggregated role statistics, and interaction-metric magnitudes are snapshot-sensitiveleak-free enriched model drops from 57.5% to 55.4%, about the same drop as the naive model; the shift re-weights familiar compositions (99.98% fall in role compositions seen on the ladder). Expert judgment was tested as a further check and proved uninformative (Section VIII): nine raters, at chance on real games above the lowest tier. A forward evaluation in an amateur league is in progress.

[3]https://draft-assistant-review.vercel.app

AI Disclosure

Portions of this manuscript's text were drafted and edited with Claude (Anthropic) [21]; all experiments, analyses, and claims were designed, verified, and approved by the authors.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] A. Kumar, A. Zhou, G. Tucker, and S. Levine, "Conservative Q-Learning for Offline Reinforcement Learning," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 1179–1191, 2020.

[2] S. Levine, A. Kumar, G. Tucker, and J. Fu, "Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems," *arXiv preprint arXiv:2005.01643*, 2020.

[3] A. Kumar, J. Fu, M. Soh, G. Tucker, and S. Levine, "Stabilizing Off-Policy Q-Learning via Bootstrapping Error Reduction," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[4] Z. Chen, Y. Sun, M. Seif El-Nasr, and T.-H. D. Nguyen, "The Art of Drafting: A Team-Oriented Hero Recommendation System for Multiplayer Online Battle Arena Games," in *Proc. ACM Conf. Recommender Systems (RecSys)*, 2018.

[5] D. Gourdeau and L. Archambault, "Discriminative Neural Network for Hero Selection in Professional Heroes of the Storm and DOTA 2," *IEEE Trans. Games*, vol. 13, no. 4, pp. 380–387, 2021.

[6] H. Lee, D. Hwang, H. Kim, B. Lee, and J. Choo, "DraftRec: Personalized Draft Recommendation for Winning in Multi-Player Online Battle Arena Games," in *Proc. ACM Web Conf. (WWW)*, 2022.

[7] A. Summerville, M. Cook, and B. Steenhuisen, "Draft-Analysis of the Ancients: Predicting Draft Picks in DotA 2 using Machine Learning," in *Proc. AAAI Conf. Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, vol. 12, no. 2, pp. 100–106, 2016.

[8] S. Chen, M. Zhu, D. Ye, W. Zhang, Q. Fu, and W. Yang, "Which Heroes to Pick? Learning to Draft in MOBA Games with Neural Networks and Tree Search," *IEEE Trans. Games*, vol. 13, no. 4, pp. 410–421, 2021.

[9] J. Lyu, X. Ma, X. Li, and Z. Lu, "Mildly Conservative Q-Learning for Offline Reinforcement Learning," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[10] I. Kostrikov, A. Nair, and S. Levine, "Offline Reinforcement Learning with Implicit Q-Learning," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[11] D. Silver *et al.*, "Mastering the Game of Go with Deep Neural Networks and Tree Search," *Nature*, vol. 529, no. 7587, pp. 484–489, 2016.

[12] D. Silver *et al.*, "A General Reinforcement Learning Algorithm that Masters Chess, Shogi, and Go through Self-Play," *Science*, vol. 362, no. 6419, pp. 1140–1144, 2018.

[13] C. Berner *et al.*, "Dota 2 with Large Scale Deep Reinforcement Learning," *arXiv preprint arXiv:1912.06680*, 2019.

[14] O. Vinyals *et al.*, "Grandmaster Level in StarCraft II using Multi-Agent Reinforcement Learning," *Nature*, vol. 575, no. 7782, pp. 350–354, 2019.

[15] S. Fujimoto and S. S. Gu, "A Minimalist Approach to Offline Reinforcement Learning," in *Advances in Neural Information Processing Systems*, vol. 34, 2021.

[16] L. Gao, J. Schulman, and J. Hilton, "Scaling Laws for Reward Model Overoptimization," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023.

[17] G. M. J.-B. Chaslot, M. H. M. Winands, and H. J. van den Herik, "Parallel Monte-Carlo Tree Search," in *Proc. Int. Conf. Computers and Games*, pp. 60–71, 2008.

[18] D. J. Wu, "Accelerating Self-Play Learning in Go," *arXiv preprint arXiv:1902.10565*, 2019.

[19] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in Data Mining: Formulation, Detection, and Avoidance," *ACM Trans. Knowledge Discovery from Data*, vol. 6, no. 4, pp. 15:1–15:21, 2012.

[20] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased Boosting with Categorical Features," in *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[21] Anthropic, "Claude (Fable 5)," large language model, 2026.